physics. Atomic physicists often make use of charged particles, electrons, accelerated by high voltage, in order to measure the binding energies of electrons. Therefore, in this case the unit of energy is that quantity of energy which is gained by an electron—or in general, by any particle which carries one elementary quantum of electric charge—in passing through a potential drop of 1 volt. This energy unit is called the *electron-volt* (*ev*). It is a very convenient unit to use when dealing with extranuclear atomic structure, for it is more or less of the same order of magnitude as the binding energies of this structure. But the binding energy of the nuclear particles is about one million times greater. Therefore, in nuclear physics it is customary to use a million times this unit—1 million ev = 1 *Mev.* One Mev is the energy which a particle carrying one elementary quantum of electricity gains on dropping through a potential difference of 1 million volts. However, it is still a very small quantity when compared with 1 erg, viz:

$$1 \text{ Mev} = 1{\cdot}6 \cdot 10^{-6} \text{ erg.}$$

We have just mentioned the energy which is liberated when a proton and a neutron combine in a deuteron. As a matter of fact, it is possible to measure this energy very well by actually producing this phenomenon. A source of neutrons is required for this purpose. Modern experimental physics has such sources available. Of course, it is true that during the process of their production, neutrons move at very high velocities and must first be slowed down in order to be brought nearly to rest, so as to be capable of combining with hydrogen atoms from this state of rest. We therefore shoot them through some substance which contains hydrogen. The neutrons collide with very many hydrogen atoms, and as a result of these collisions, they gradually lose most of their kinetic energy, except for a very small quantity which corresponds to the temperature of that particular substance. They end up with a so-called thermal velocity. In this state they are caused to combine with protons.

In this process, the binding energy of the deuteron is liberated, and in conformity with the law of the conservation of energy, it must still exist somewhere—in other words, it must go somewhere, in some form. The most logical assumption is that it leaves in the form of an electromagnetic radiation, namely, as gamma radiation of an extremely short wavelength. It is therefore to be expected that the formation of each deuteron, from one proton and one neutron, is associated with the emission of a gamma-ray photon very